Friday, June 12, 2026 8:34 PM

DECISION TREES – 2 ⇒

Agenda →
1. Recap
2. Example →
3. gini impurity
4. Bias variance
5. Concepts ⇒

ENTROPY ⇒ Randomness
↳ impurity

Goal → Homogeneous Regions

DATA

i/p features ⇒ go play tennis or not

Training Data

Outlook	Temperature	Humidity	Wind	Tennis
sunny	hot	high	weak	no
sunny	hot	high	strong	no
overcast	hot	high	weak	yes
rain	mild	high	weak	yes
rain	cool	normal	weak	yes
rain	cool	normal	strong	no
overcast	cool	normal	strong	yes
sunny	mild	high	weak	no
sunny	cool	normal	weak	yes
rain	mild	normal	weak	yes
sunny	mild	normal	strong	yes
overcast	mild	high	strong	yes
overcast	hot	normal	weak	yes
rain	mild	high	strong	no

$n_{yes} = 9$

$n_{no} = 5$

14 dp

$H(Y) = 1$

n_{yes} n_{no} → 9 dp

Pred^n ⇒ YES ⇒ =

50 - 50

Entropy of Root Node

Entropy of Root Node

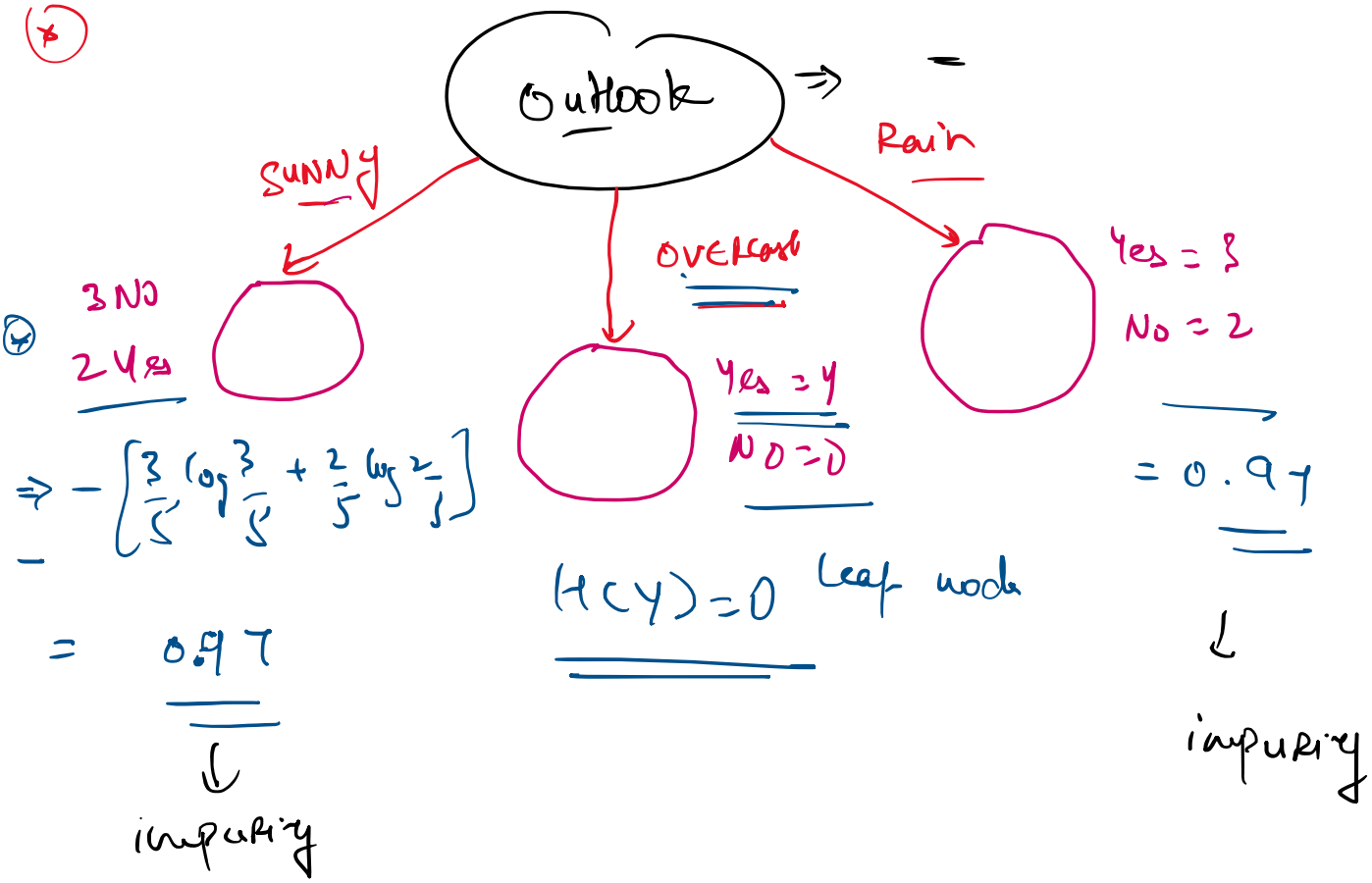
$$H = -\left[\frac{9}{14}\log\frac{9}{14} + \frac{5}{14}\log\frac{5}{14}\right]$$

log2 ↑ np.log2 ↑

$$= 0.94$$

Feature to split on

→ max info gain / min entropy



(4) info gain = H(Parent) − H(children)

$$= 0.94 - \left[\frac{5}{14} \times 0.97 + \frac{0+5}{14} \times 0.97\right]$$

$$= 0.94 - 0.692$$

$$= 0.24$$ info gain

Reduction in Entropy

⇒ Calculate for other features as well

IG (outlook) = 0.24

(temp) = ?

(Humidity) = ?

max. ⇓ info

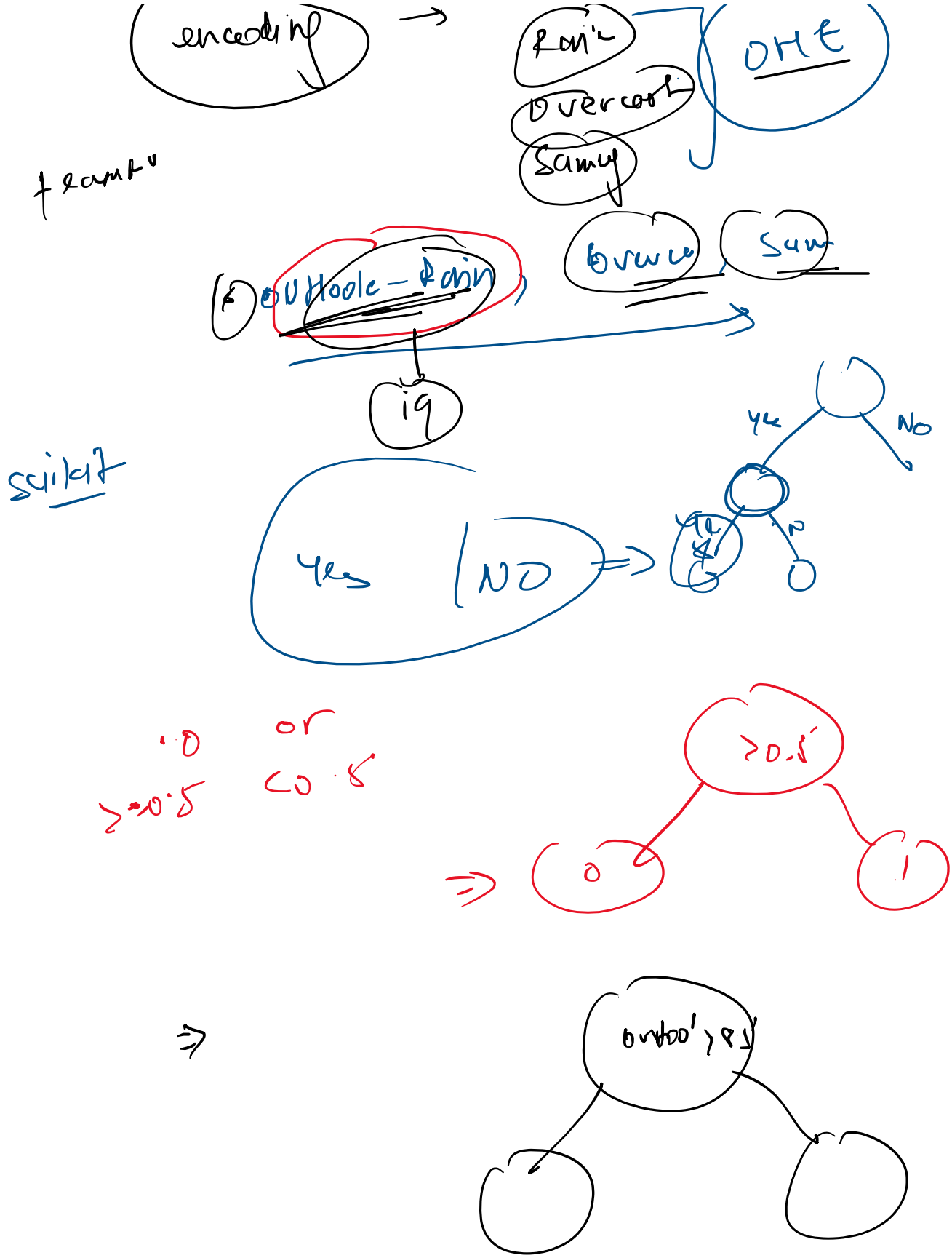
select

(5) Categorical data ⇒

encoding →

outlook → sklearn

rain → OHE


encoding
Rain
Overcast
Sunny
OHE
Outlook – Rain
Overcast
Sunny
iq
scikit
Yes
No
Yes
No
>0.5
0
1

⇒ * Numerical features ⇒

<

* gini impurity ⇒ another way to compute impurity

entropy = $-\sum_{i=1}^{k} p_i \cdot \log p_i$

heavy

expensive

* gini impurity = $1 - \sum_{i=1}^{k} [P(y_i)]^2$

$P(y_i) \geq P(y_i)$

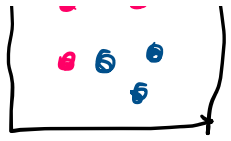
* $H(Y) = 1$

$H(Y)$

1o - 1o

* $GI(S_1) =$

$\Rightarrow 1 - [P(Red)^2 + P(Blue)^2]$



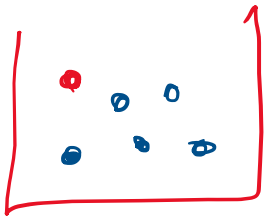
S_1

$$\Rightarrow 1 - \left[P(\text{Red})^2 + P(\text{Blue})^2 \right]$$

$$= 1 - \left[(0.5)^2 + (0.5)^2 \right]$$

$$= 1 - \left[\underline{0.25} + \underline{0.25} \right]$$

$$= 0.5$$

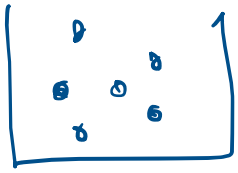


S_2

$$GI(S_2) = 1 - \left[\left(\frac{5}{6}\right)^2 + \left(\frac{1}{6}\right)^2 \right]$$

$$= 1 - \text{---}$$

$$= 0.27$$



S_3

$$GI(S_3) = 1 - \left[\left(\frac{0}{6}\right)^2 + \left(\frac{6}{6}\right)^2 \right]$$

$$= 1 - 1 = 0$$

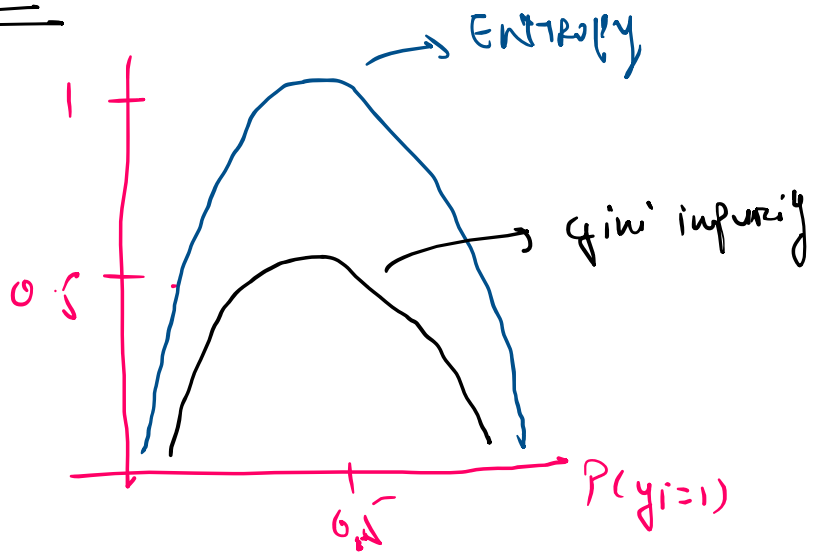
entropy $\Rightarrow [0, 1]$

gini $\Rightarrow [0, 0.5]$

$\Rightarrow$ 2 classes

gini

(How ⇒ k classes)



criterion = 'entropy' ✓

Sklearn → gini impurity

60 → A 40 → B

60/100 = 0

$= 1 - \{(0.6)^2 + (0.4)^2\}$

$0.36 + 0.16$

0.48

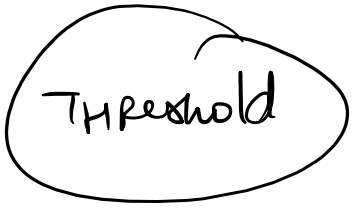
Numerical features ⇒

Categorical

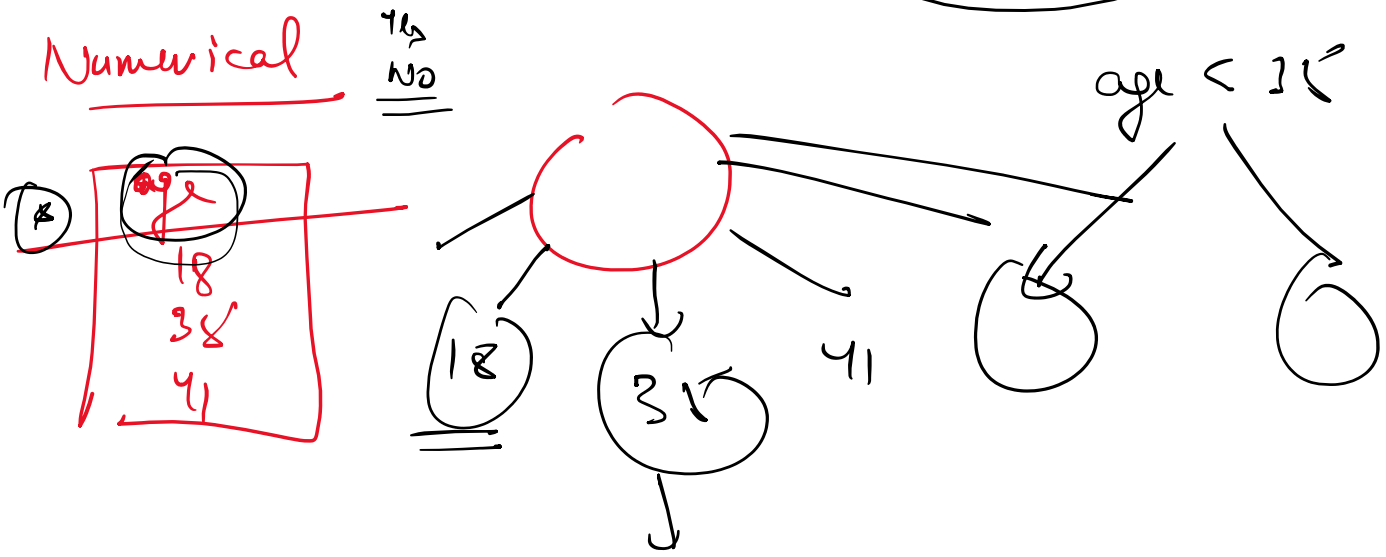
3

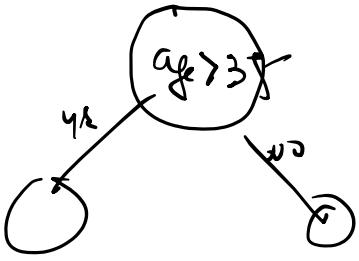
2



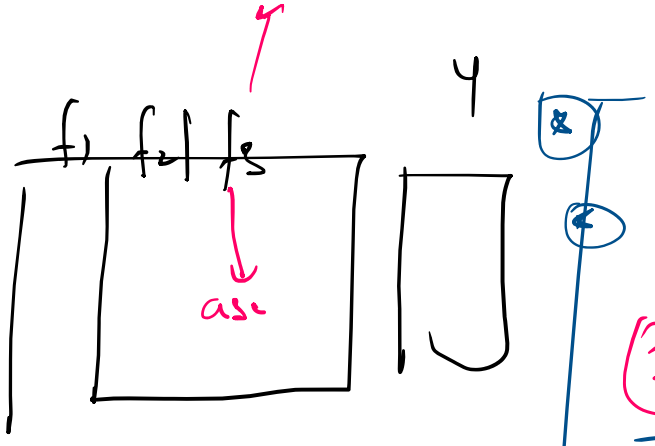


Numerical





Numerical





Step (1) → arrange asc. order

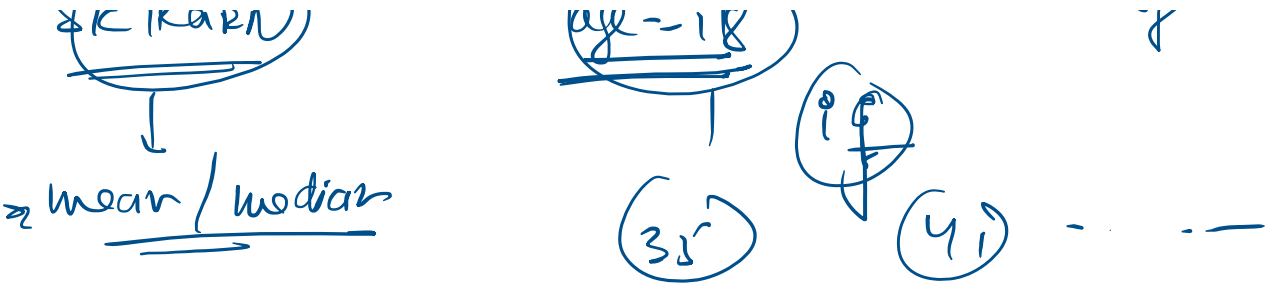
(2) Take each unique value (n1, n2, ..., nn) as threshold & calculate IG

age > ? No

sklearn

age - 18

age - 12



=> Computationally expensive

* 18, 25, 31 - - - -

* Binning / Bucketization

EDA → Buckets

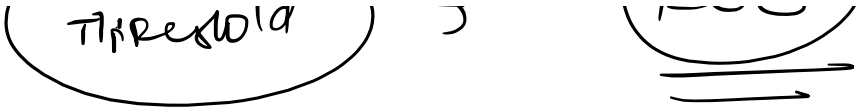
age → {(18-25) (25-35) (35-50) (50-60)}

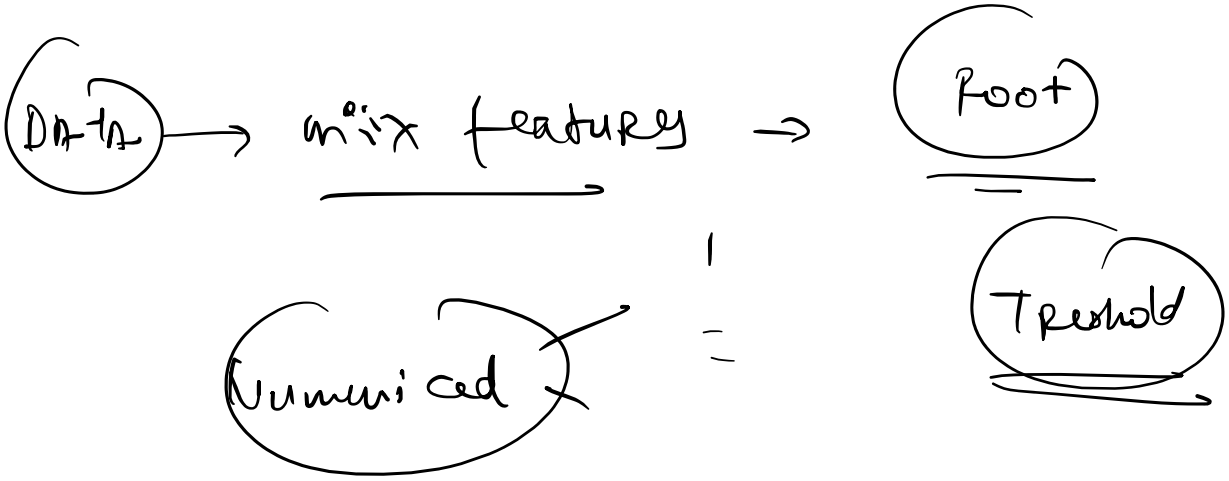
* Categorical ⇒ ⇒ Categorical

L2(ig)

* Sklearn

* Threshold → ROC





Sklearn → mean | median

Underfit | overfit

Underfit →

Too simple

depth ↓

⇒ High Bias

OVERFIT ⇒

Too complex

depth ↑

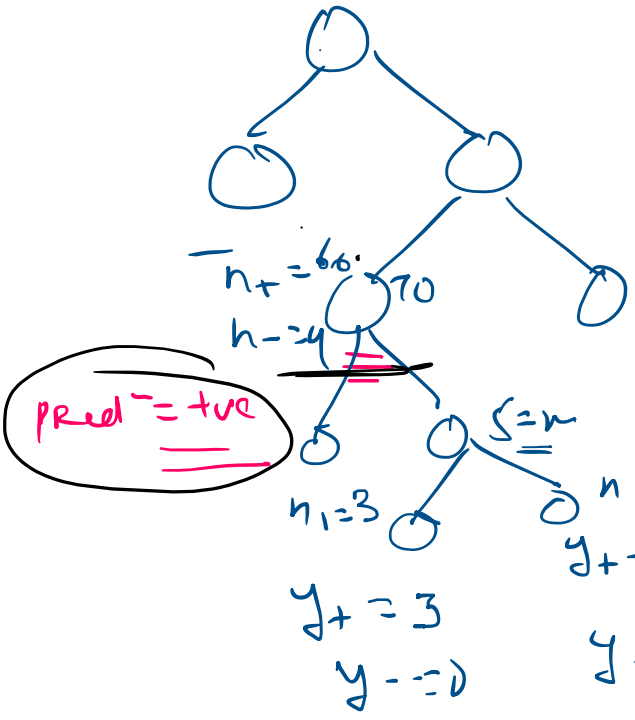
low Bias &

High VAR.

⇒ [illegible]

→ low VAR.

High VAR.



Pred^n = -ve

Post

↳ Build → overfitting

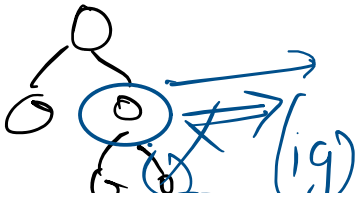
Pruning

Pruning

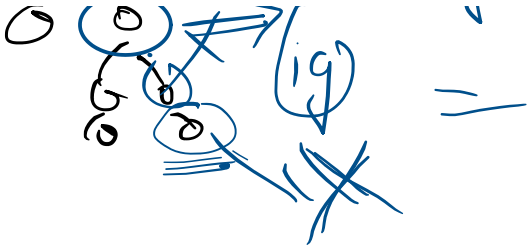
Pre Pruning

Eg

Cutting irRelevant Branches



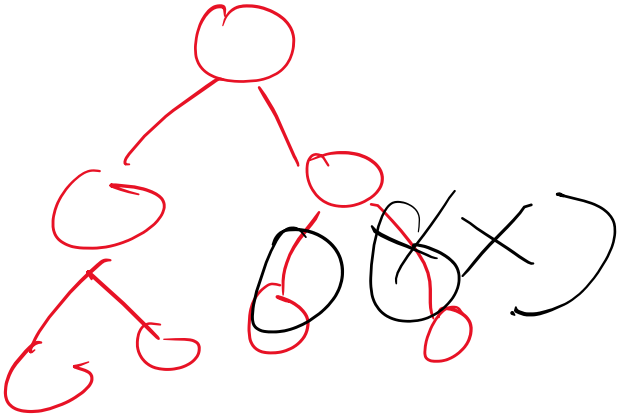
if gets impure



Bagging

α → ccp-alpha

↳ Regularization → tree



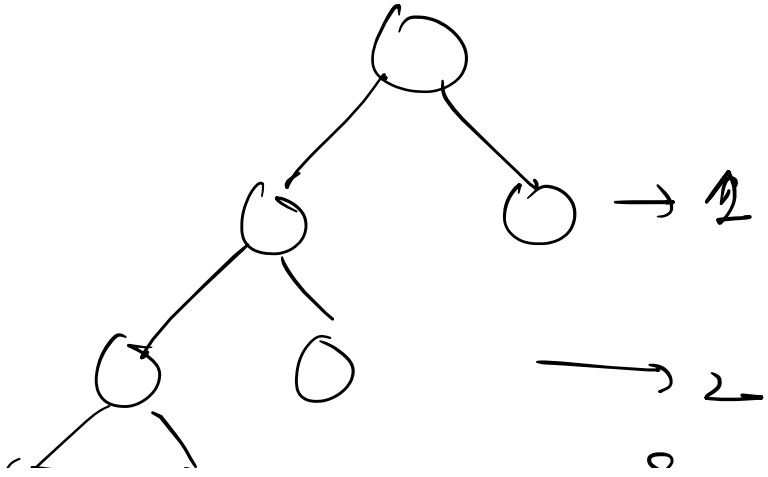
(*) less gust = Better nodes

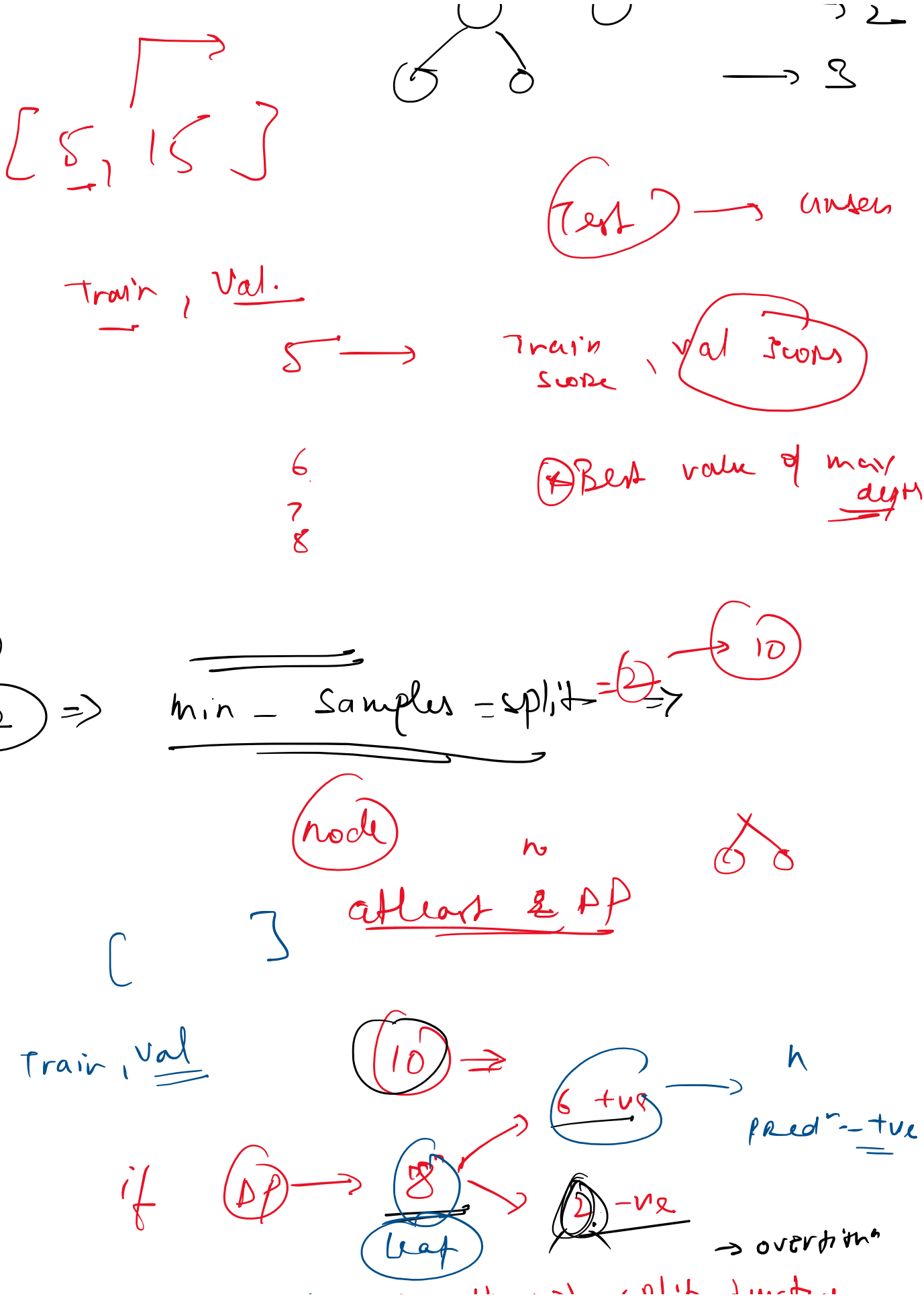
(*)

(*) Hyperparam tuning

(1) max-depth

↳ Hyperparam tuning

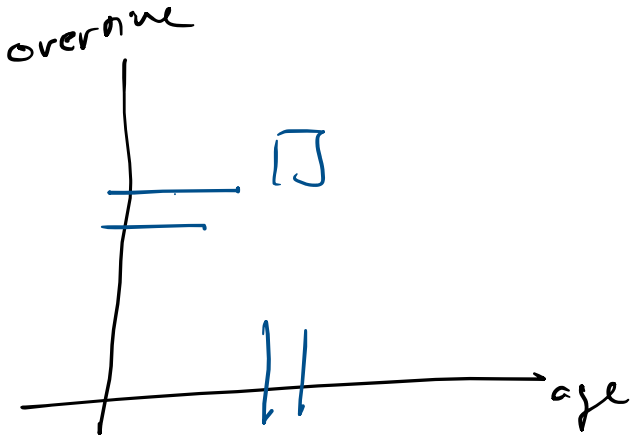



Train , Val.
Train score , val scores
6
7
8
Best value of max depth
min_samples_split
node
atleast 2 DP
Train , val
10
6 +ve
pred^n -- +ve
if
DP
8
leaf
2 -ve
overfitting

(leaf) → overfitting

(*) Hyperparam Tuning → will not split further

ENTROPY ;



(10)

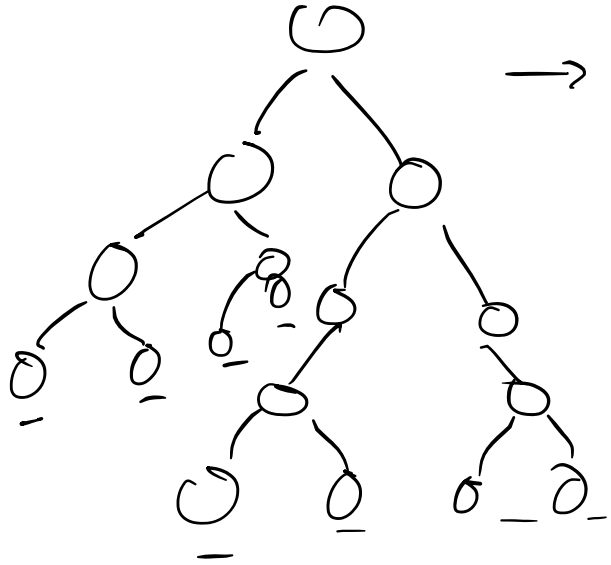
10-15 ⇒ Reduced the overfitting

Hyper

(50)

(*) max_leaf_node ⇒ max no of leaf nodes to a ward

↳ node



→ Hyperparam Tuning

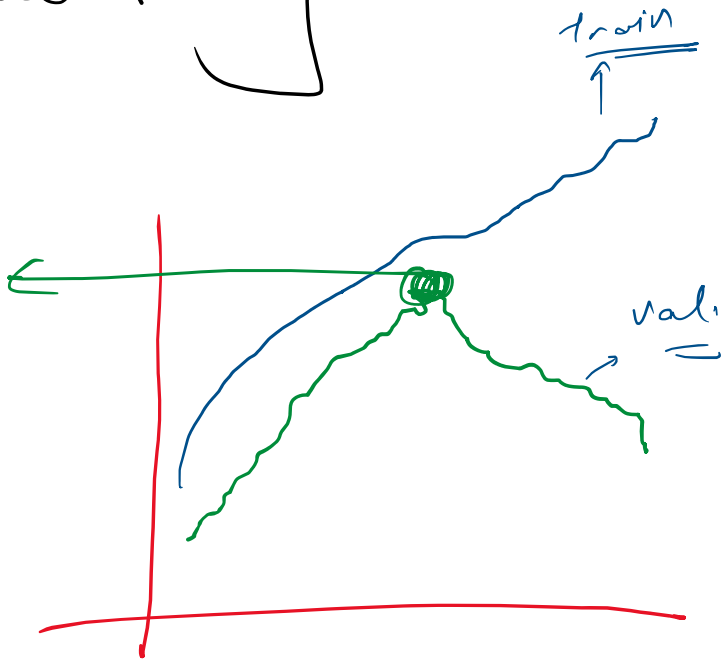
⇒ Overfitting

1. max_depth
2. min_samples_leaf
3. min_samples_split
4. max_features
5. max_leaf_nodes

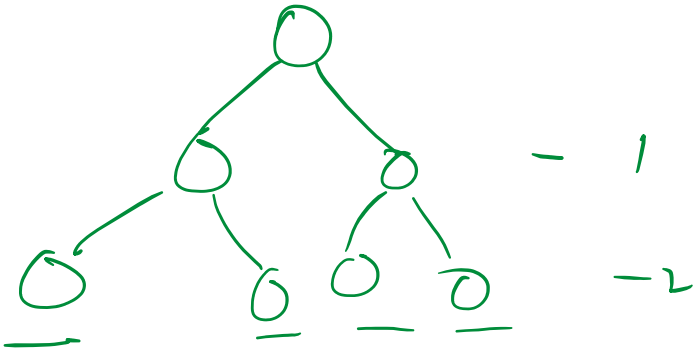
[ ]

Tune these

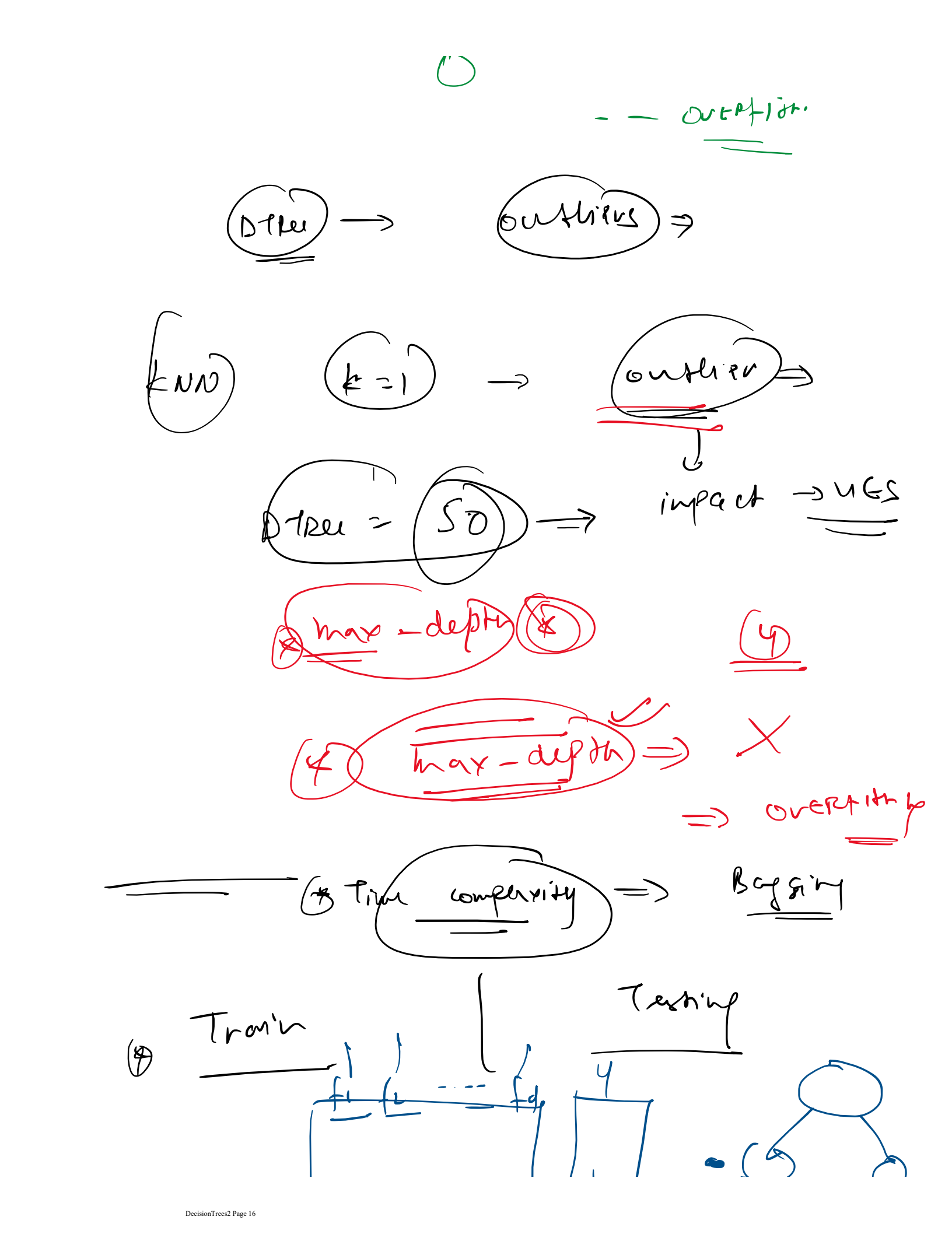
Val - acc

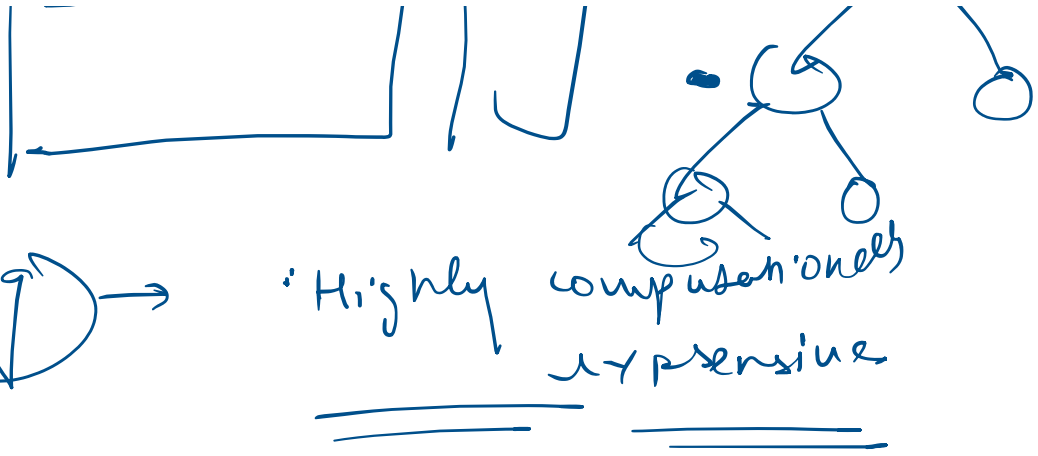


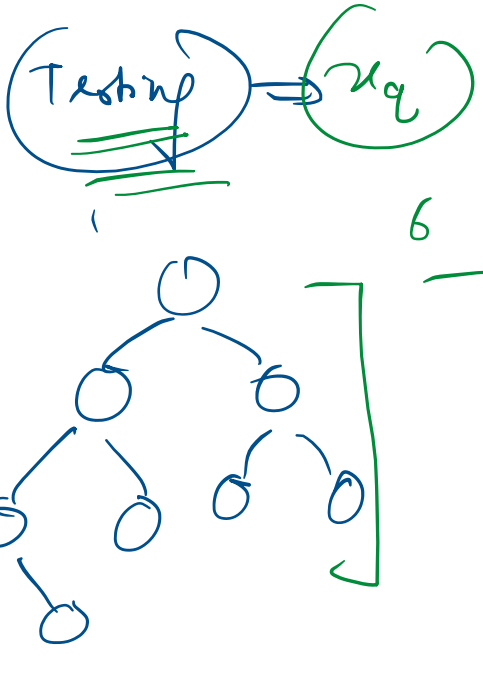
depth = 2



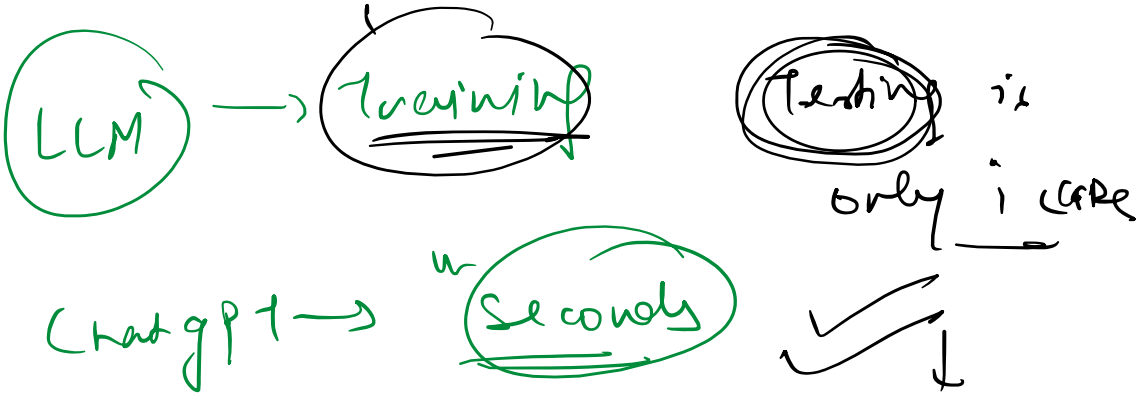
Too simple

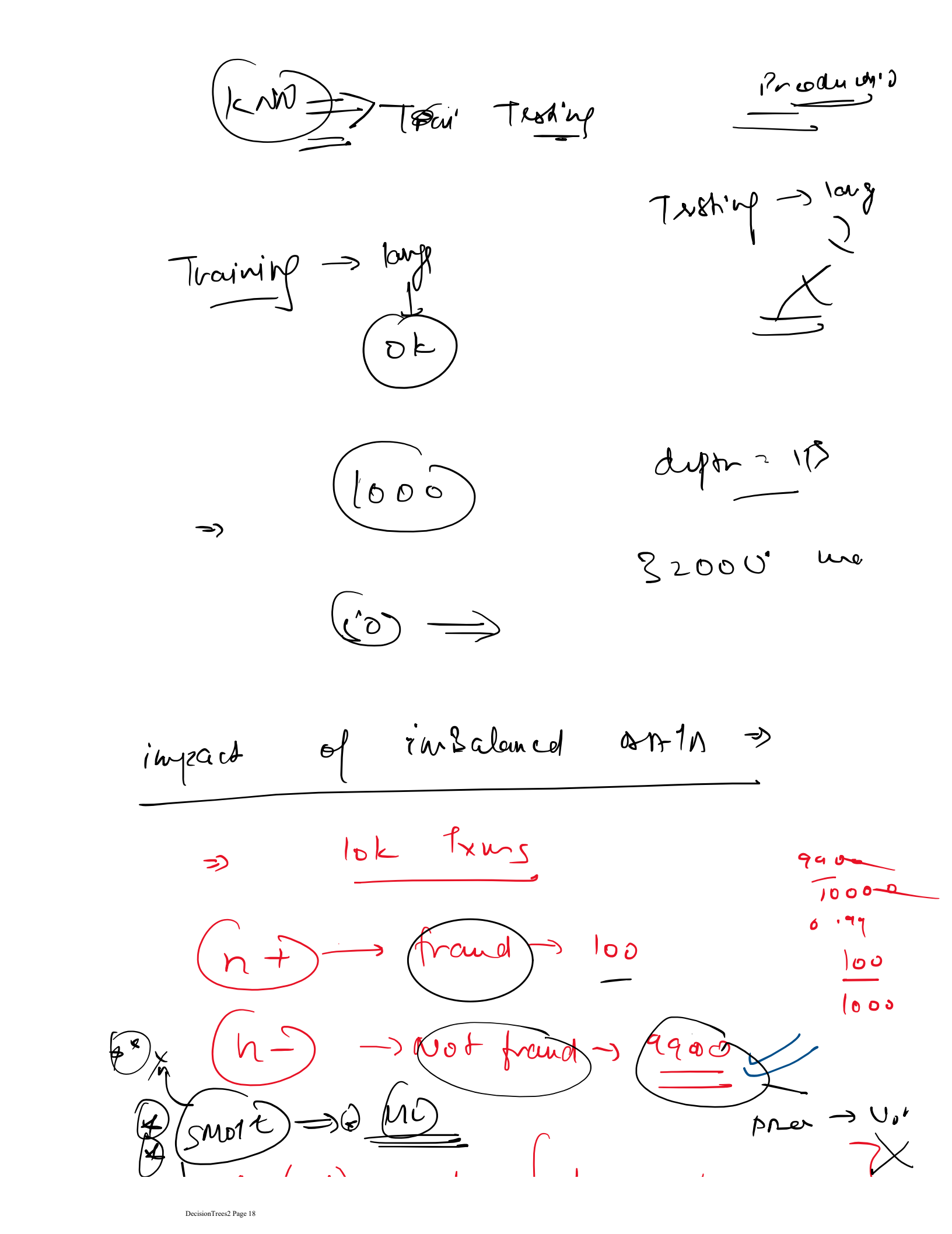

overfitting
DTree → outliers ⇒
KNN
k=1 → outlier ⇒
impact → LGS
DTree = SO ⇒
max-depth
max-depth ⇒ X
⇒ overfitting
Time complexity ⇒ Bagging
Training
Testing


Highly computationally expensive


Testing
if else cond
O(6)
depth = 1
depth = 2
O(depth)

⇒ Production


LLM
Training
Testing is only i care
ChatGPT
Secondly


KNN
Testing
Training
large
ok
Testing
large
1000
depth = 15
2000
one
impact of imbalanced data
10k txns
n+
fraud
100
n-
not fraud
9900
SMOTE
ML
prev

SMOTE

$gi(y) = 1 - [(0.99)^2 - \quad]$

$= 0.019$

good

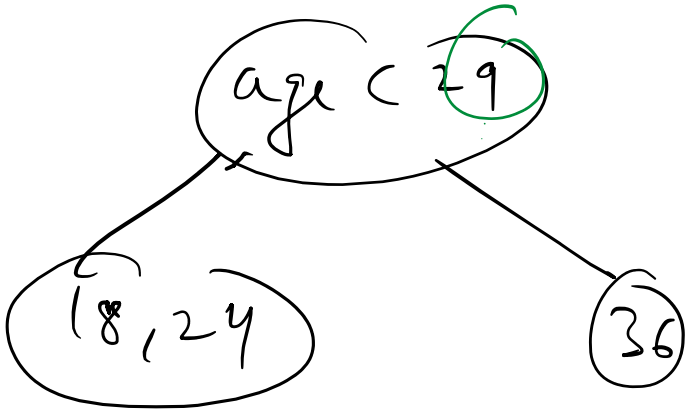
Syllabus

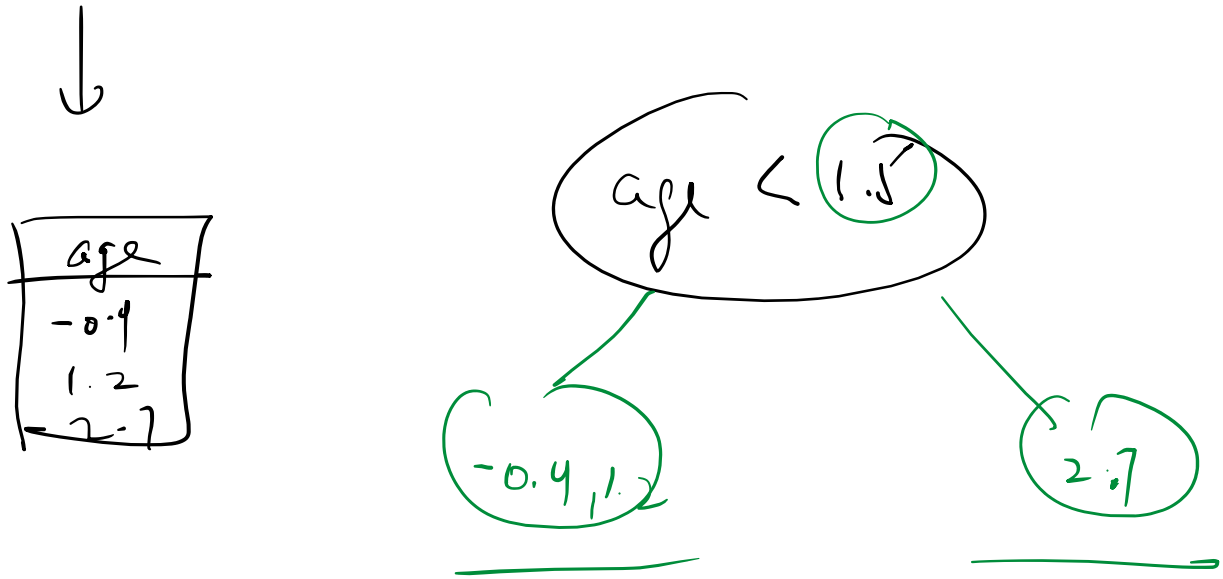
$\Rightarrow$ Impact of imbalancement

Reduce this impact $\rightarrow$ SMOTE / class weight

DTrees $\rightarrow$ Require Scaling?

age
18
24
39





Not Required → Scaling

Hack → KNN, Linear Reg

1 - 1000 [1-10] → distances

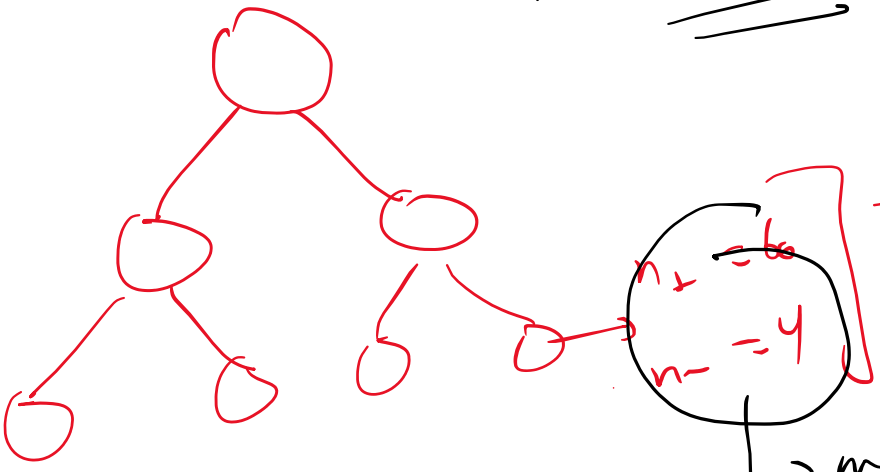
data → similar scale

DTrees → no distance →

⑥ Decision Trees Regression ⇒

Tv → cat.

↳ Numerical

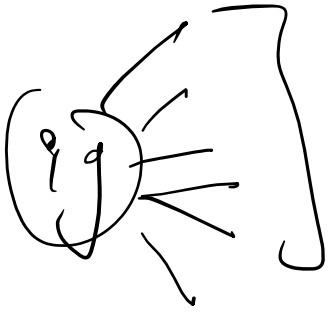


→ majority

↳ mean / median → pred

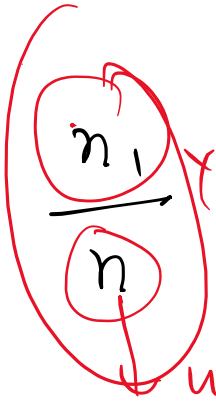
⑥ ⇒ ⑧ Feature imp ⇒ Next class

⇒



max → 1 imp feature

Root — DT



$\times I_G(F_i)$ → [ ]

⑧ → Feature imp ⇒